Client Retention Automation System

An intelligent platform that automatically re-engages fitness studio clients through personalized follow-up emails and one-click booking links.

Built for FitLife Studio — Fitness Industry

System Documentation & Technical Explainer

March 2026 • Version 1.0

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1. The Problem

Why fitness studios lose clients — and revenue — silently

Fitness studios face a critical challenge: **client drop-off is invisible**. When a client stops booking sessions, there's no alarm bell. Days turn into weeks, weeks into months, and by the time anyone notices, the client has moved on.

67%	3-7	5x	$0
CLIENTS WHO LAPSE NEVER RETURN	DAY WINDOW TO RE-ENGAGE	CHEAPER TO RETAIN THAN ACQUIRE	REVENUE FROM A LAPSED CLIENT

The Manual Approach Fails

Most studios rely on staff to manually track who hasn't visited recently and send follow-up messages. This approach breaks down because:

- ✓ Staff are busy with in-person clients and can't track every inactive member
- ✓ Follow-ups are inconsistent — some clients get contacted, others slip through
- ✓ No personalization — generic "we miss you" messages feel impersonal
- ✓ No direct booking link — clients have to navigate the website themselves
- ✓ No visibility into which clients are at risk until it's too late

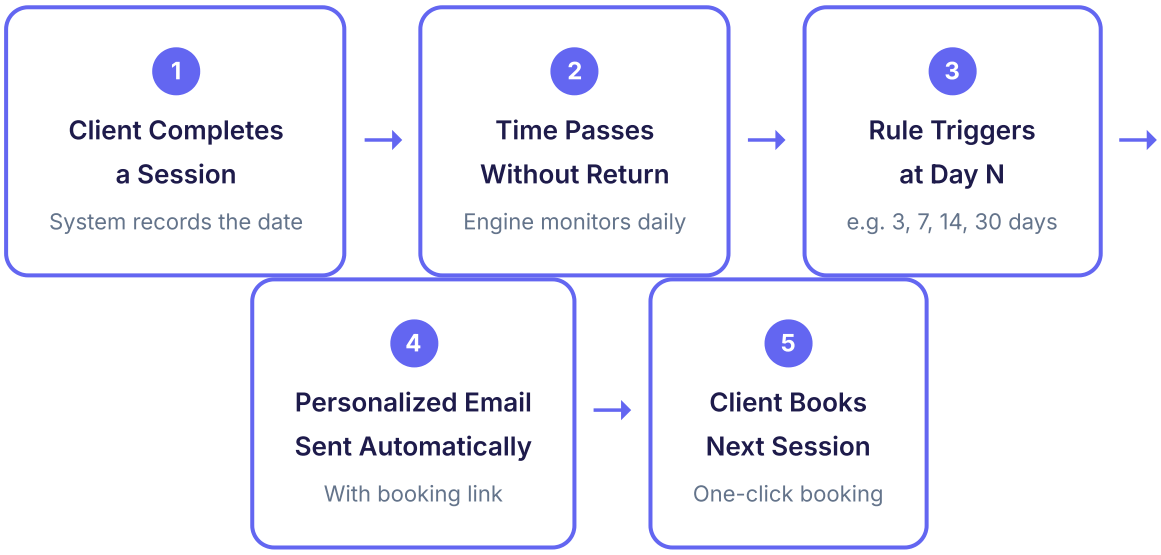
> **The core insight:** The window to re-engage a lapsed client is narrow. A follow-up on day 3 has 4x the success rate of a follow-up on day 30. Automation ensures no client falls through the cracks.

2. The Solution

Automated retention that works while you train

The Client Retention Automation System continuously monitors your client roster and automatically sends personalized follow-up emails at the right intervals — each with a direct booking link that lets clients schedule their next session in one click.

How It Works — The Core Flow



What Makes It Different

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Time-Aware Triggers

Four configurable rules fire at exactly the right intervals — day 3 (gentle nudge), day 7 (check-in), day 14 (special offer), day 30 (win-back).

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Personalized Messages

Each email uses the client's name, session history, and studio branding. Templates use {{placeholder}} variables for dynamic content.

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One-Click Booking Links

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Real-Time Dashboard

Every email contains a unique, time-limited booking link. Clients pick a date and session type — done. No login required.

See at-risk clients, email delivery status, booking conversion rates, and follow-up activity — all in one professional interface.

3. System Architecture

Clean, modular design built for reliability

Dashboard UI — Single-page app (Tailwind CSS + Vanilla JS)

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REST API Layer — Express.js with 17+ endpoints across 6 route modules

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Service Layer — Retention Engine • Email Service • Booking Service • Scheduler

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Data Layer — SQLite (better-sqlite3) • 6 tables • Prepared statements

Database Schema

Table	Purpose	Key Fields
clients	Client roster with contact info and status	name, email, phone, status, membership_type, last_session_at
sessions	Every training session logged	client_id, session_date, session_type, duration_minutes, completed
follow_ups	Every automated outreach record	client_id, rule_id, status (pending/sent/failed), sent_at
bookings	Unique booking links with token tracking	client_id, token (UUID), expires_at, used (0/1)

retention_rules	Configurable follow-up trigger rules	name, trigger_days, priority, enabled, template_id
email_templates	Email content with dynamic placeholders	name, subject, body (with {{variables}})

4. Core Features

Everything a studio owner needs to retain clients automatically

Client Management

Add clients via the dashboard or API. Each client record tracks name, email, phone, membership type (monthly/annual/drop-in), join date, and current status (active/at-risk/inactive). The system automatically updates a client's status based on session activity.

Session Tracking

Every completed session is logged with date, type (Yoga, HIIT, Pilates, Strength, etc.), duration, and notes. The system uses the most recent completed session to calculate "days since last visit" — the key metric that drives retention triggers.

Automated Follow-Up Emails

The retention engine runs on a schedule (hourly in production, every 10 seconds in demo mode). It scans all active clients, checks if any match a retention rule based on their days-since-last-session, and sends personalized emails automatically.

Each email includes:

- ✓ Client's first name and personalized greeting
- ✓ Studio branding (name, colors, professional layout)
- ✓ A clear call-to-action button with a unique booking link
- ✓ The booking link expires after 72 hours for urgency
- ✓ Professional HTML layout with responsive design

One-Click Booking

When a client clicks their booking link, they see a branded booking page where they can: select a session date, choose a session type, and confirm — all without logging in. The system creates the session, marks the booking as used, and updates the client's last_session_at timestamp.

Duplicate Prevention

Smart deduplication ensures a client won't receive the same follow-up rule twice. If a client already has a pending follow-up for a given rule, the system skips it. This prevents email fatigue and keeps communication respectful.

5. The Retention Engine

The brain of the system — rule-based, configurable, and smart

Default Retention Rules

Rule	Trigger	Priority	Email Tone
Gentle Nudge	3 days since last session	Low	Friendly check-in, "Ready for your next session?"
Check-In	7 days since last session	Medium	Concerned, "We've noticed you haven't visited"
Special Offer	14 days since last session	High	Incentive-driven, "Special offer just for you"
Win-Back	30 days since last session	Critical	Re-engagement, "We miss you at the studio"

Engine Logic (Simplified)

```
// Every hour (or 10s in demo mode), the engine:

1. Queries all active clients with their days_since_last_session
2. For each client, finds matching retention rules (days_since >= trigger_days)
3. Picks the highest-priority matching rule
4. Checks if a follow-up already exists for this client + rule
5. If no duplicate: creates booking link + sends email + records follow-up
6. Logs everything for the dashboard to display
```

Email Template System

Templates use `{{placeholder}}` syntax for dynamic content. Available variables:

Variable	Value
`{{name}}`	Client's full name

`{{first_name}}`	Client's first name
`{{email}}`	Client's email address
`{{days_since}}`	Days since last session
`{{booking_link}}`	Unique booking URL
`{{studio_name}}`	Studio name from config

6. Dashboard Overview

A professional SaaS-style control center for studio owners

The dashboard is a single-page web application accessible at `http://localhost:3000`. It provides real-time visibility into client activity, follow-up status, and retention metrics.

Dashboard Sections

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Stats Overview

5 key metrics at a glance: total clients, active clients, at-risk count, follow-ups sent, and booking conversion rate.

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At-Risk Clients

Highlighted panel showing clients who haven't booked recently, sorted by days since last session. Quick actions: record session, create booking link.

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Client Management

Full client table with search and filter. Click any client to see their complete session history, follow-up timeline, and quick actions.

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Follow-Up Activity

Live feed of all follow-up emails — pending, sent, and failed. Each entry links to the Ethereal email preview for verification.

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Retention Rules

View and toggle retention rules. Visual funnel showing how many clients are at each stage (3-day, 7-day, 14-day, 30-day).

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Demo Controls

Time acceleration panel: advance the clock by N days to simulate client lapse. Trigger manual retention checks. View all sent email previews. Reset demo data.

Interactive Features

- ✓ **Add Client** — Form modal to register new clients with all fields
- ✓ **Record Session** — Quick-action button on any client to log a new session
- ✓ **Create Booking Link** — Generate a unique booking URL for any client
- ✓ **Toggle Rules** — Enable/disable retention rules with one click
- ✓ **Auto-Refresh** — Dashboard updates every 5 seconds automatically
- ✓ **Toast Notifications** — Success/error feedback for every action

7. API Reference

RESTful API with 17+ endpoints across 6 modules

Method	Endpoint	Description
Clients		
GET	/api/clients	List all clients
GET	/api/clients/:id	Get client details
POST	/api/clients	Add a new client
GET	/api/clients/:id/sessions	Client's session history
GET	/api/clients/:id/follow-ups	Client's follow-up history
Sessions		
GET	/api/sessions	List all sessions
POST	/api/sessions	Record a new session
POST	/api/sessions/:id/complete	Mark session as completed
Follow-Ups		
GET	/api/follow-ups	List all follow-ups
GET	/api/follow-ups/pending	Pending follow-ups only
POST	/api/follow-ups/trigger	Manually trigger retention check
Bookings		
GET	/api/bookings	List all bookings
GET	/api/bookings/:token	Get booking by token
POST	/api/bookings/create/:clientId	Create booking link

POST	/api/bookings/:token/confirm	Confirm a booking
Dashboard		
GET	/api/dashboard/stats	Aggregated metrics
GET	/api/dashboard/at-risk	At-risk client list
GET	/api/dashboard/rules	Retention rules

All endpoints return a consistent `{ success: true, data: ... }` response envelope. Errors return `{ success: false, error: "message" }` with appropriate HTTP status codes.

8. Demo Walkthrough

Step-by-step guide to showcase the system

Setup (30 seconds)

```
# Install dependencies and seed demo data
npm install
npm run seed

# Start in demo mode (scheduler runs every 10 seconds)
DEMO_MODE=true npm run dev

# Open dashboard
open http://localhost:3000
```

Demo Script

Act 1 — Show the Dashboard (2 min)

- ✓ Point out the 5 key metrics at the top — especially "At-Risk Clients"
- ✓ Click on an at-risk client to show their profile and session gap
- ✓ Show the follow-up activity feed — emails already sent automatically
- ✓ Click an email preview link to show the actual branded email

Act 2 — Demonstrate Time Acceleration (3 min)

- ✓ Go to the Demo Controls section
- ✓ Click "Advance +3 Days" — watch new clients become at-risk
- ✓ Click "Trigger Retention Check" — watch emails fire in real-time
- ✓ Show the follow-up feed updating with new sent emails
- ✓ Click an email preview to show the personalized content and booking link

Act 3 — Show the Booking Flow (2 min)

- ✓ Copy a booking link from an email preview

- ✓ Open it in a new tab — show the branded booking page
- ✓ Select a date and session type, click "Confirm Booking"
- ✓ Return to dashboard — show the client is now active again

Act 4 — Studio Owner Actions (1 min)

- ✓ Add a new client using the "Add Client" button
- ✓ Record a session for an existing client
- ✓ Toggle a retention rule on/off
- ✓ Show how everything updates in real-time

9. Tech Stack & Setup

Production-ready technologies with zero external dependencies

Technology Choices

Component	Technology	Why
Runtime	Node.js 20+	Fast, widely supported, great for I/O-heavy workloads
Web Framework	Express.js	Industry standard, minimal overhead, huge ecosystem
Database	SQLite (better-sqlite3)	Zero config, portable, synchronous API, perfect for single-server
Email	Nodemailer + Ethereal	Real SMTP delivery in production; test accounts for demo
Scheduler	node-cron	Lightweight, in-process cron for recurring retention checks
Frontend	Tailwind CSS + Vanilla JS	No build step, fast loading, professional UI out of the box
Booking Tokens	UUID v4	Cryptographically random, collision-proof, URL-safe

Quick Start

```
# Clone the repository
git clone https://github.com/98soumithr/client-retention-fitness.git
cd client-retention-fitness

# Install dependencies
npm install

# Seed demo data (15 clients, 149 sessions, 4 rules, 4 templates)
npm run seed

# Start the server
```

```
npm run dev

# Open in browser
open http://localhost:3000
```

Environment Variables

Variable	Default	Description
PORT	3000	Server port
BASE_URL	http://localhost:3000	Base URL for booking links
DEMO_MODE	false	Enable 10-second scheduler interval

10. Future Roadmap

Where this system can grow

SMS Notifications

Add Twilio integration for SMS follow-ups alongside email, with preference management per client.

AI-Powered Timing

Machine learning model to predict optimal follow-up timing per client based on historical patterns.

Payment Integration

Stripe integration so booking links can also collect payment for paid sessions or packages.

Calendar Sync

Google Calendar / Outlook integration to add confirmed bookings directly to client calendars.

Mobile App

React Native companion app for studio owners to manage retention on the go.

Multi-Location

Support for franchise/multi-location studios with per-location rules and dashboards.

This is a fully functional v1.0 — it works today, handles real email delivery, and provides a complete demo experience. The roadmap items above are enhancements that can be added incrementally as the product grows.

Ready to retain more clients?

github.com/98soumithr/client-retention-fitness

Built with Node.js, Express, SQLite • March 2026